



**2008**

**TRIAL HIGHER SCHOOL CERTIFICATE  
EXAMINATION**

# Mathematics

## **General Instructions**

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

## **Total Marks – 120**

- Attempt Questions 1 – 10
- All questions are of equal value

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**Total Marks – 120****Attempt Questions 1 – 10****All questions are of equal value**

Begin each question in a SEPARATE writing booklet. Extra booklets are available.

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| <b>Question 1</b> (12 marks) Use a SEPARATE writing booklet.                                 | <b>Marks</b> |
|----------------------------------------------------------------------------------------------|--------------|
| a) Evaluate $\left(\frac{1}{e^{2.5}} - 1\right)^2$ correct to 3 significant figures.         | <b>2</b>     |
| b) Solve $ 2x - 1  \leq 3$                                                                   | <b>2</b>     |
| c) If $\frac{4}{2 - \sqrt{3}} = a + b\sqrt{3}$ find the values of $a$ and $b$ .              | <b>2</b>     |
| d) Find the tenth term of the arithmetic series<br>$4\frac{1}{2} + 3 + 1\frac{1}{2} + \dots$ | <b>2</b>     |
| e) Factorise $2x^2 + 7x - 4$                                                                 | <b>2</b>     |
| f) Find the perpendicular distance from the point $(1, 3)$ to the line<br>$6x - 8y + 5 = 0$  | <b>2</b>     |

**End of Question 1**

**Question 2** (12 marks) Use a SEPARATE writing booklet.**Marks**a) Differentiate with respect to  $x$ 

(i)  $x \sin x$  **2**

(ii)  $(1 + e^x)^5$  **2**

b) (i) Find  $\int \sec^2 3x \, dx$  **2**

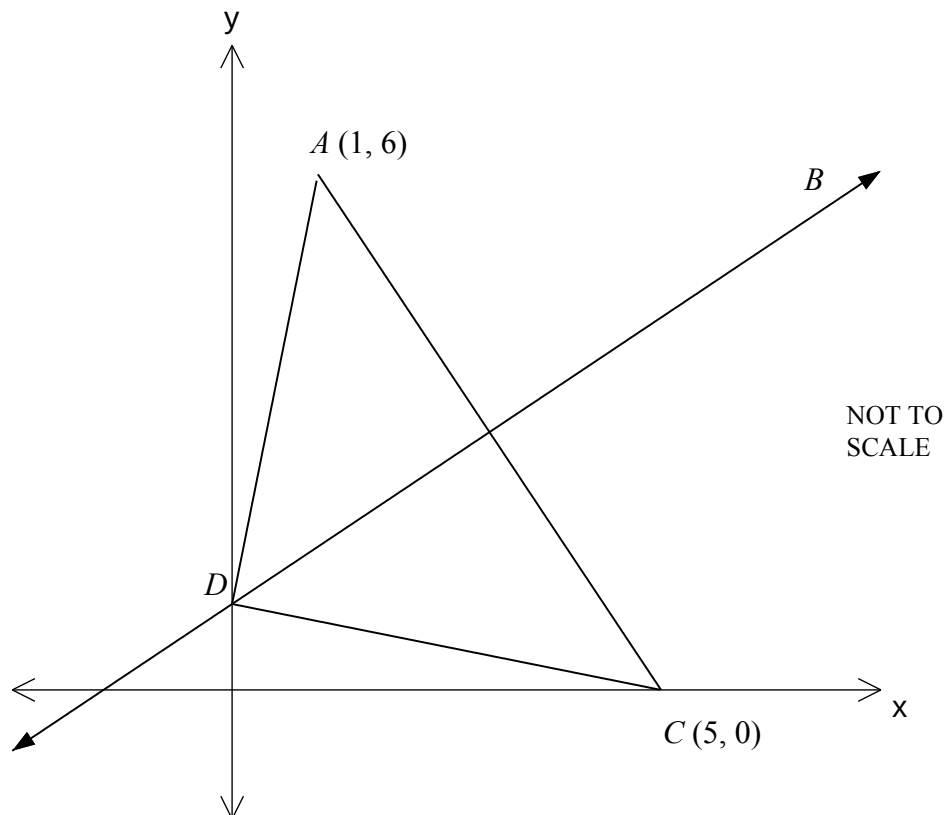
(ii) Evaluate  $\int_0^3 \frac{6x}{1+x^2} \, dx$  **3**

c) Find the equation of the tangent to the curve  $y = \cos 3x$  at the point whose  $x$ -coordinate is  $\frac{\pi}{6}$ . **3**

**End of Question 2**

**Question 3** (12 marks) Use a SEPARATE writing booklet.**Marks**

a)



The points  $A$  and  $C$  have coordinates  $(1, 6)$  and  $(5, 0)$  respectively.  
The line  $BD$  has an equation of  $2x - 3y + 3 = 0$  and meets the  $y$  axis in  $D$ .

- |      |                                                                                    |   |
|------|------------------------------------------------------------------------------------|---|
| i)   | The point $M$ is the midpoint of $AC$ . Show that $M$ has coordinates $(3, 3)$ .   | 1 |
| ii)  | Show that $M$ lies on $BD$ .                                                       | 1 |
| iii) | Find the gradient of the line $AC$ .                                               | 1 |
| iv)  | Show that $BD$ is perpendicular to $AC$ .                                          | 2 |
| v)   | Find the distance $AC$ .                                                           | 1 |
| vi)  | Explain why the quadrilateral $ABCD$ is a kite regardless of the position of $B$ . | 1 |

**Question 3 continues on next page.**

**Question 3 continued****Marks**

- b) Michael is training for a local marathon. He has trained by completing practice runs over the marathon course. So far he has completed three practice runs with times shown below.

| Week 1  | Week 2             | Week 3                        |
|---------|--------------------|-------------------------------|
| 3 hours | 2 hours 51 minutes | 2 hours 42 minutes 27 seconds |

- i) Show that these times form a geometric series with a common ratio  $r = 0.95$ . **1**
- ii) If this series continues, what would be his expected time in Week 5, to the nearest second? **1**
- iii) How many hours, minutes and seconds (to the nearest second ) will he have run, in total, in his practice runs in these 5 weeks? **1**
- iv) If the previous winning time for the marathon was 2 hours and 6 minutes, how many weeks must he keep practicing to be able to run the marathon in less than the previous winning time? **2**

**End of Question 3**

**Question 4** (12 marks) Use a SEPARATE writing booklet.**Marks**

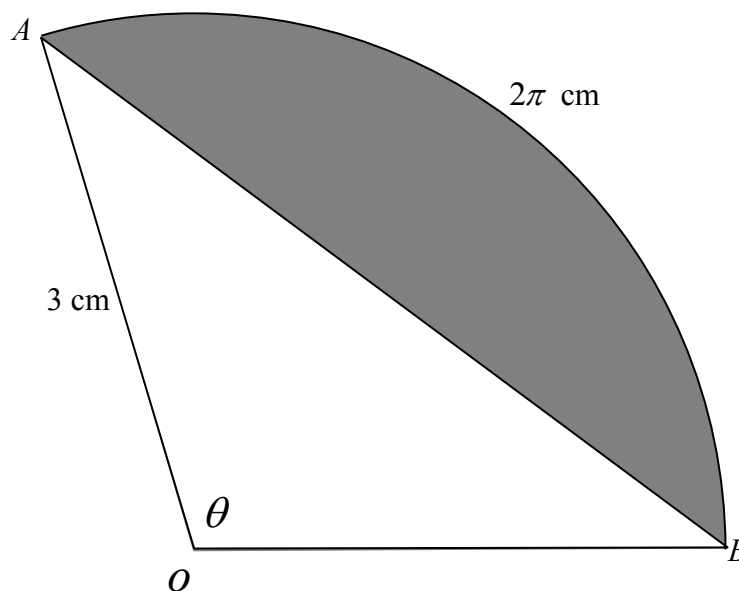
a) Show that:

**2**

$$\sqrt{\frac{\operatorname{cosec}^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}} = \tan x$$

b) For the function  $y = x^3 - 3x^2 - 9x + 1$ , find the values of  $x$  for which the curve is decreasing.**3**

c)



$AOB$  is a sector of a circle, centre  $O$  and radius 3 cm.  
The length of arc  $AB$  is  $2\pi$  cm.  $AB$  is a chord.

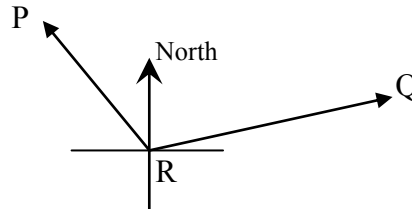
(i) Calculate the angle  $\theta$  subtended at the centre of the circle by the arc  $AB$ .**1**

(ii) Calculate the exact area of the shaded segment.

**2****Question 4 continues on next page.**

**Question 4 continued**

- d) Peta and Quentin are pilots of two light planes which leave Resthaven station at the same time. Peta flies on a bearing of  $330^\circ$  at a speed of 180 km/h and Quentin flies on a bearing of  $080^\circ$  at a speed of 240 km/h. Copy the diagram below onto your answer page and mark the information on the diagram.

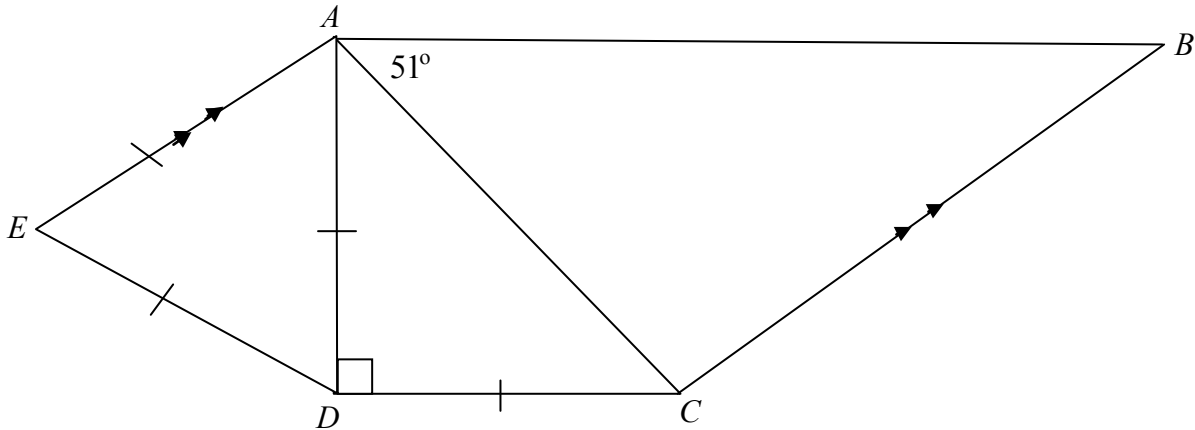


- |     |                                                                                         |          |
|-----|-----------------------------------------------------------------------------------------|----------|
| i)  | Show that Peta and Quentin are 692 km (to the nearest km) apart after 2 hours?          | <b>2</b> |
| ii) | What is the bearing of Quentin from Peta after 2 hours. (Answer to the nearest degree.) | <b>2</b> |

**End of Question 4**

**Question 5** (12 marks) Use a SEPARATE writing booklet.**Marks**

- a) In the diagram below  $AE = ED = AD = DC$ ,  $\angle ADC = 90^\circ$  and  $AE \parallel BC$ .  
 $\angle BAC = 51^\circ$



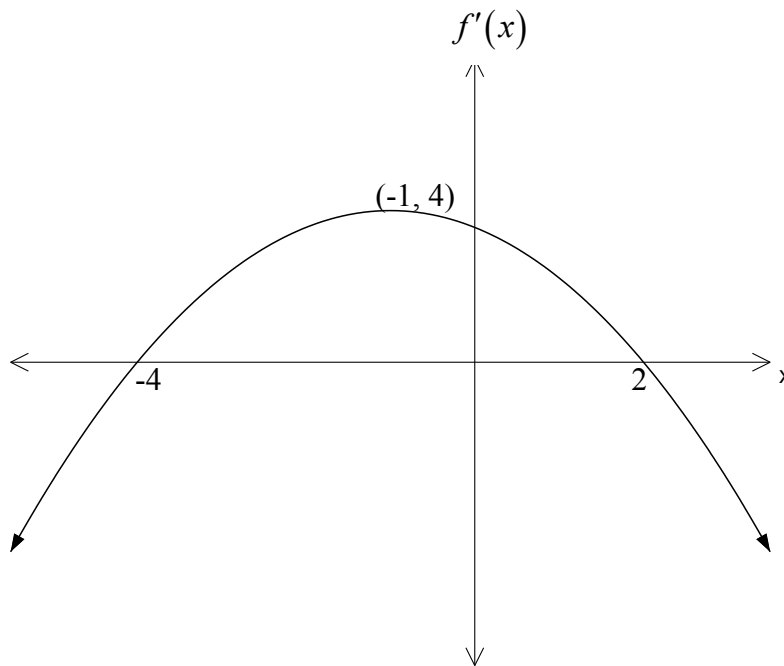
- i) Find the size of  $\angle EAB$ . Give reasons for your answer. **3**
- ii) Find the size of  $\angle ABC$ . Give reasons for your answer. **1**
- b) A particle moves in a straight line so that its displacement, in metres, is given by  

$$x = \frac{4t^2 + t + 8}{4t + 1}$$
 where  $t$  is measured in seconds.
- i) Find the initial displacement of the particle. **1**
- ii) Find an expression for the velocity of the particle. **1**
- iii) Show that the particle is stationary when  $t = \frac{-1 + 4\sqrt{2}}{4}$  seconds (or  
 1.2 seconds to one decimal place). **2**
- iv) Find the total distance travelled in the first two seconds. (Answer to 1  
 decimal place.) **2**
- c) Solve the pair of simultaneous equations **2**
- $$3x - y = 10$$
- $$x = y + 2$$

**End of Question 5**

**Question 6** (12 marks) Use a SEPARATE writing booklet.**Marks**a) For the function  $y = x^6 - 6x^4$ 

- i) Find the  $x$  coordinates of the points where the curve crosses the axes. **2**
- ii) Find the coordinates of the stationary points and determine their nature. **4**
- iii) Find the coordinates of any points of inflexion. **2**
- iv) Sketch the graph of  $y = x^6 - 6x^4$  indicating clearly the intercepts, stationary points and points of inflexion. **2**

b) For a certain function  $y = f(x)$ , the sketch of  $y = f'(x)$  is shown.

Give the  $x$  coordinates of the stationary points on  $y = f(x)$  and indicate if they are maxima or minima. **2**

**End of Question 6**

**Question 7** (12 marks) Use a SEPARATE writing booklet.**Marks**

a) Let  $\log_a 2 = x$  and  $\log_a 5 = y$ . Find an expression, in terms of  $x$  and  $y$ , for

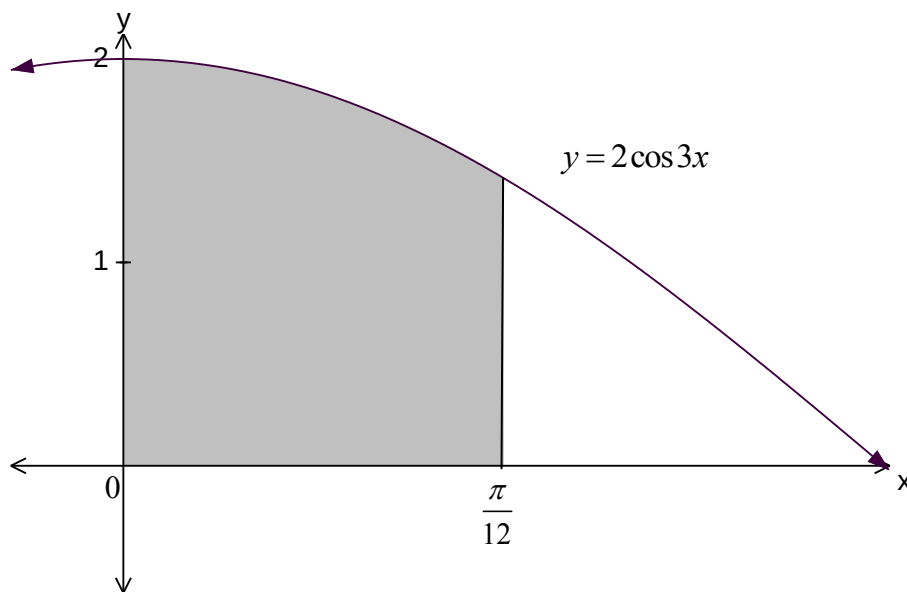
i)  $\log_a 0.4$

**1**

ii)  $\log_a 20$

**2**

b)



The diagram shows the graph of  $y = 2 \cos 3x$ .

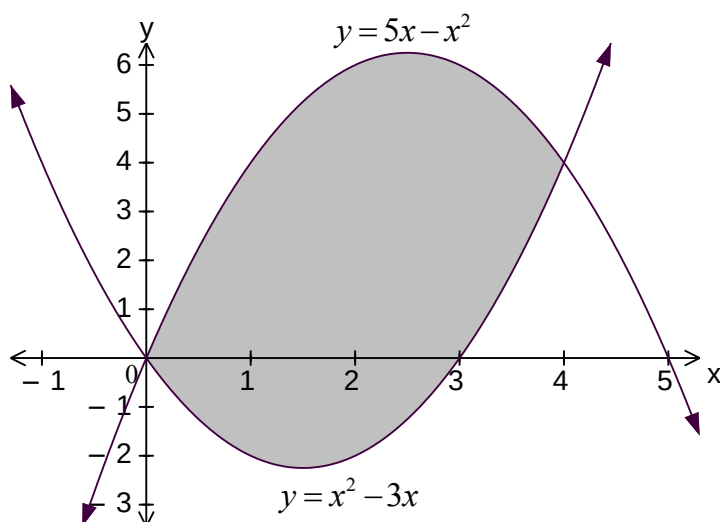
**3**

Find the area enclosed by the curve  $y = 2 \cos 3x$ , the line  $x = \frac{\pi}{12}$  and the  $x$  and  $y$  axes.

**Question 7 continues on next page.**

**Question 7 continued**

c)



- i) Show that the curves  $y = x^2 - 3x$  and  $y = 5x - x^2$  intersect at the points  $(0, 0)$  and  $(4, 4)$ . 2
- ii) Find the area enclosed between the two curves. 3
- d) Find  $\frac{d}{dx} \log_e (\cos x)$ . 1

**End of Question 7**

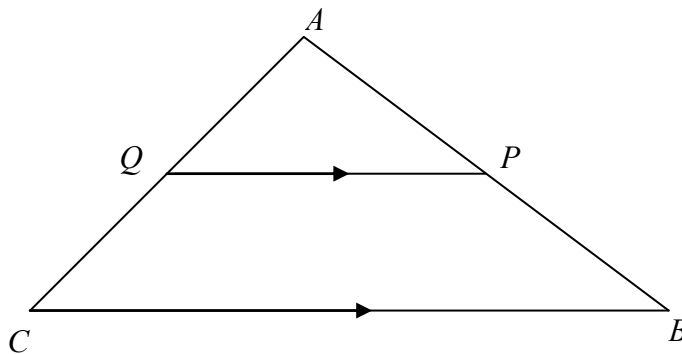
**Question 8** (12 marks) Use a SEPARATE writing booklet.**Marks**

- a) A city has a population which is growing at a rate that is proportional to the current population. The population at time  $t$  years is given by

$$P = Ae^{kt}$$

- i) Show that  $P = Ae^{kt}$  satisfies the equation  $\frac{dP}{dt} = kP$ . 1
- ii) If the population at the start of 2006 when  $t = 1$  was 147 200 and at the start of 2007 when  $t = 2$  was 154 800, find the values of  $A$  and  $k$ . 2
- iii) Find the population at the start of 2009. 1
- iv) Find during which year the population will first exceed 200 000. 1

- b) In the diagram below,  $P$  is the midpoint of the side  $AB$  of the  $\triangle ABC$ .  $PQ$  is drawn parallel to  $BC$ .



- i) Prove that  $\triangle ABC \parallel \triangle APQ$ . 2
- ii) Explain why  $Q$  is the midpoint of  $AC$ . 2
- c) Find an approximation for  $\int_1^3 g(x) dx$  by using Simpson's Rule with the values in the table below. 2

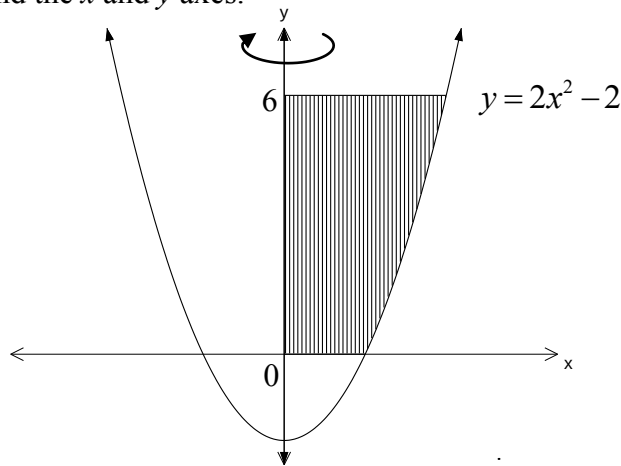
|        |    |     |   |     |   |
|--------|----|-----|---|-----|---|
| $x$    | 1  | 1.5 | 2 | 2.5 | 3 |
| $g(x)$ | 12 | 8   | 0 | 3   | 5 |

- d) Evaluate  $\sum_{n=2}^5 (n^2 - 1)$  1

**End of Question 8**

**Question 9** (12 marks) Use a SEPARATE writing booklet.**Marks**

- a) The diagram shows the region bounded by the curve  $y = 2x^2 - 2$  the line  $y = 6$  and the  $x$  and  $y$  axes.

**3**

Find the volume of the solid of revolution formed when the region is rotated about the  $y$  axis.

- b) Sketch the function  $y = \ln(x+1)$ , showing its essential features.

**2**

- c) Given that  $\frac{d}{dx}(e^{x^5}) = 5x^4 e^{x^5}$ , find  $\int 2x^4 e^{x^5} dx$ .

**1**

- d) A car dealership has a car for sale for a cash price of \$20 000. It can also be bought on terms over three years. The first six months are interest free and after that interest is charged at the rate of 1% per month on that months balance. Repayments are to be made in equal monthly instalments beginning at the end of the first month.

A customer buys the car on these terms and agrees to monthly repayments of \$ $M$ . Let \$ $A_n$  be the amount owing at the end of the  $n$ th month.

- Find an expression for  $A_6$ . **1**
- Show that  $A_8 = (20\,000 - 6M)1.01^2 - M(1 + \square + \square 1.01)$  **1**
- Find an expression for  $A_{36}$ . **2**
- Find the value of  $M$ . **2**

**End of Question 9**

**Question 10** (12 marks) Use a SEPARATE writing booklet.**Marks**

- a) A plant nursery has a watering system which repeatedly fills a storage tank then empties its contents to water different sections of the nursery. The volume of water (in cubic metres) in the tank at a time  $t$  is given by the equation

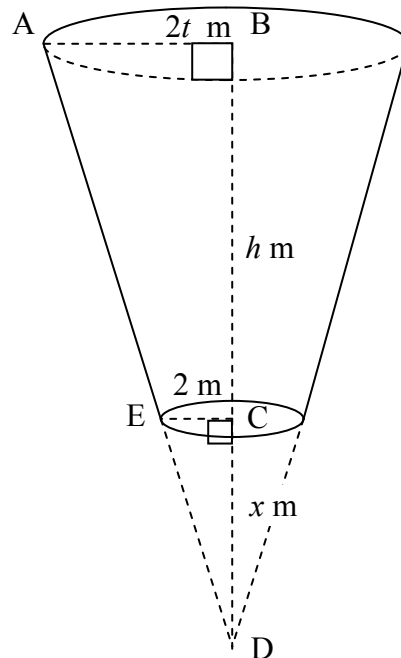
$$V = 2 - \sqrt{3} \cos t - \sin t \text{ where } t \text{ is measured in minutes.}$$

- |      |                                                                                             |          |
|------|---------------------------------------------------------------------------------------------|----------|
| i)   | Give an equation for $\frac{dV}{dt}$ , the rate of change of the volume at a time $t$ .     | <b>1</b> |
| ii)  | Is the tank initially filling or emptying?                                                  | <b>1</b> |
| iii) | At what time does the tank first become completely full and what is its capacity when full? | <b>3</b> |

**Question 10 continues on next page.**

**Question 10 continued****Marks**

- b) A truncated cone is to be used as a part of a hopper for a grain harvester. It has a total height of  $h$  metres. The top radius is to be  $t$  times greater than the bottom radius which is 2 metres.



$AB = 2t$  metres  
 $BC = h$  metres  
 $EC = 2$  metres  
 $CD = x$  metres

- i) If  $x$  is the height of the removed section of the original cone, show using similar triangles that  $x = \frac{h}{t-1}$  **2**
- ii) Show that the volume of the truncated cone is given by  $V = \left(\frac{4\pi h}{3}\right)(t^2 + t + 1)$  **2**
- iii) If the upper radius plus the lower radius plus the height of the truncated cone must total 12 metres, calculate the maximum volume of the hopper. **3**

**End of Examination**

**STANDARD INTEGRALS**

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x, \quad x > 0$

2008  
TRIAL HSC  
EXAMINATION

# Mathematics

## **SOLUTIONS**

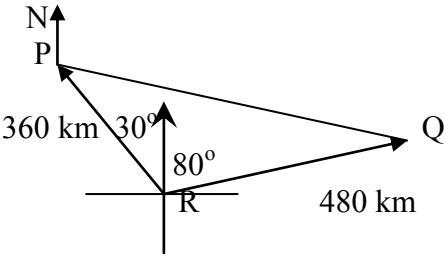
| Question 1 |                                                                                                                                                                                                                                            | Trial HSC Examination- Mathematics |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                   | Marks                              |
| (a)        | $\left(\frac{1}{e^{2.5}} - 1\right)^2 = 0.84256 = 0.843 \text{ (3 sig fig)}$                                                                                                                                                               | 2                                  |
| (b)        | $ 2x - 1  \leq 3$<br>$-3 \leq 2x - 1 \leq 3$<br>$-2 \leq 2x \leq 4$<br>$-1 \leq x \leq 2$                                                                                                                                                  | 2                                  |
| (c)        | $\frac{4}{2 - \sqrt{3}} = a + b\sqrt{3}$<br>$\frac{4}{2 - \sqrt{3}} = \frac{4}{2 - \sqrt{3}} \times \frac{2 + \sqrt{3}}{2 + \sqrt{3}}$<br>$= \frac{8 + 4\sqrt{3}}{4 - 3}$<br>$a + b\sqrt{3} = 8 + 4\sqrt{3}$<br>$a = 8 \text{ and } b = 4$ | 2                                  |
| (d)        | $4\frac{1}{2} + 3 + 1\frac{1}{2} + \dots$<br>Series is arithmetic with $a = 4\frac{1}{2}$ and $d = -1\frac{1}{2}$<br>$T_n = a + (n - 1)d$<br>$T_{10} = 4\frac{1}{2} + 9 \times -1\frac{1}{2}$<br>$T_{10} = -9$                             | 2                                  |
| (e)        | $(x + 4)(2x - 1)$                                                                                                                                                                                                                          | 2                                  |
| (f)        | $d = \frac{ 6(1) - 8(3) + 5 }{\sqrt{6^2 + (-8)^2}}$<br>$= \frac{ -13 }{\sqrt{100}}$<br>$= 1.3 \text{ units}$                                                                                                                               | 2                                  |

| Question 2 |                                                   | Trial HSC Examination- Mathematics |
|------------|---------------------------------------------------|------------------------------------|
| Part       | Solution                                          | Marks                              |
| (a) i)     | $x \cos x + \sin x$                               | 2                                  |
| ii)        | $5(1 + e^x)^4 \times e^x$<br>$= 5e^x (1 + e^x)^4$ | 2                                  |
| (b) i)     | $\frac{1}{3} \tan 3x + c$                         | 2                                  |

| Question 2 |                                                                                                                                                                                                                                                                                                                          | Trial HSC Examination- Mathematics |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                 | Marks                              |
| ii)        | $\left[ 3 \ln(1+x^2) \right]_0^3$ $= 3[\ln 10 - \ln 1]$ $= 3 \ln 10$                                                                                                                                                                                                                                                     | 3                                  |
| (c)        | $y' = -3 \sin 3x$ <p>when <math>x = \frac{\pi}{6}</math>, <math>y' = -3 \sin \frac{\pi}{2}</math></p> $= -3$ <p><math>\therefore</math> gradient of tangent is <math>-3</math></p> <p>when <math>x = \frac{\pi}{6}</math>, <math>y = 0</math>.</p> <p>Equation of tangent is <math>3x + y - \frac{\pi}{2} = 0</math></p> | 3                                  |

| Question 3 |                                                                                                                                                                                                                                                                               | Trial HSC Examination- Mathematics |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                      | Marks                              |
| (a)i)      | <p>Midpoint of (1, 6) and (5, 0).</p> $MP = \left( \frac{1+5}{2}, \frac{6+0}{2} \right) = \left( \frac{6}{2}, \frac{6}{2} \right) = (3, 3)$                                                                                                                                   | 1                                  |
| ii)        | <p>Show that (3,3) lies on <math>2x - 3y + 3 = 0</math></p> $LHS = 2(3) - 3(3) + 3$ $= 6 - 9 + 3$ $= 0 = RHS$ <p>So M lies on BD.</p>                                                                                                                                         | 1                                  |
| iii)       | <p>Gradient AC = <math>m_1 = \frac{6-0}{1-5} = \frac{6}{-4} = -\frac{3}{2}</math></p>                                                                                                                                                                                         | 1                                  |
| iv)        | <p>Find gradient <math>m_2</math> of BD <math>2x - 3y + 3 = 0</math></p> $2x - 3y + 3 = 0$ $3y = 2x + 3$ $y = \frac{2}{3}x + 1$ $\therefore m_2 = \frac{2}{3}$ $m_1 \cdot m_2 = -\frac{3}{2} \cdot \frac{2}{3} = -1$ <p><math>\therefore</math> BD is perpendicular to AC</p> | 2                                  |

| Question 3 |                                                                                                                                                                                                                                                                                                                                                       | Trial HSC Examination- Mathematics |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                                              | Marks                              |
| v)         | $AC = \sqrt{(5-1)^2 + (0-6)^2}$ $= \sqrt{16+36}$ $= \sqrt{52}$ $= 2\sqrt{13} \text{ units}$                                                                                                                                                                                                                                                           | 1                                  |
| (vi)       | <p>The lines AC and BD would form the diagonals of the quadrilateral ABCD.<br/> BD is the perpendicular bisector of AC from ii and iv above.<br/> The diagonals of a kite meet at right angles and one diagonal bisects the other,<br/> so ABCD meets the criteria for a kite.</p>                                                                    | 1                                  |
| (b)i)      | $\frac{2 \text{ hours } 51 \text{ minutes}}{3 \text{ hours}} = \frac{2.85}{3} = 0.95$ $\frac{2 \text{ hours } 42 \text{ minutes } 27 \text{ seconds}}{2 \text{ hours } 51 \text{ minutes}} = \frac{2.7075}{2.85} = 0.95$ <p><math>\therefore</math> forms a geometric series with <math>r = 0.95</math></p>                                           | 1                                  |
| ii)        | $a = 3, n = 5 \text{ and } r = 0.95$<br>$u_5 = ar^{n-1}$<br>$= 3 \times 0.95^4$<br>$= 2.4435...$<br>$= 2 \text{ hours } 26 \text{ min } 37 \text{ s}$                                                                                                                                                                                                 | 1                                  |
| iii)       | $s_n = \frac{a(1-r^n)}{(1-r)}$ $= \frac{3(1-0.95^5)}{(1-0.95)}$ $= 13.573...$ $= 13 \text{ hours } 34 \text{ min } 23 \text{ s (to nearest s)}$                                                                                                                                                                                                       | 1                                  |
| iv)        | $u_n = ar^{n-1}$<br>$2 \text{ hours } 6 \text{ min} = 3 \times 0.95^{n-1}$<br>$3 \times 0.95^{n-1} = 2.1$<br>$0.95^{n-1} = 2.1 \div 3 = 0.7$<br>$\ln(0.95^{n-1}) = \ln(0.7)$<br>$(n-1)\ln(0.95) = \ln(0.7)$<br>$n-1 = \frac{\ln(0.7)}{\ln(0.95)}$<br>$n-1 = 6.953$<br>$n = 7.953...$<br><p>Would need to continue for 8 weeks to better the time.</p> | 2                                  |

| Question 4 |                                                                                                                                                                                                                                                                                                               | Trial HSC Examination- Mathematics |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                      | Marks                              |
| (a)        | $\sqrt{\frac{\operatorname{cosec}^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}} = \sqrt{\frac{1 - \cos^2 x}{\cos^2 x}}$ $= \sqrt{\frac{\sin^2 x}{\cos^2 x}}$ $= \frac{\sin x}{\cos x}$ $= \tan x$                                                                                                                     | 2                                  |
| (b)        | $y' = 3x^2 - 6x - 9$ $x^2 - 2x - 3 < 0$ $(x+1)(x-3) < 0$ $\therefore \text{function is decreasing for } -1 < x < 3$                                                                                                                                                                                           | 3                                  |
| (c)i)      | $l = r\theta$ $2\pi = 3\theta$ $\theta = \frac{2\pi}{3}$                                                                                                                                                                                                                                                      | 1                                  |
| (c)ii)     | $A = \frac{1}{2}r^2(\theta - \sin \theta)$ $A = \frac{1}{2} \times 3^2 \left( \frac{2\pi}{3} - \sin \frac{2\pi}{3} \right)$ $A = \frac{9}{2} \left( \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right)$ $A = \frac{9}{2} \left( \frac{4\pi - 3\sqrt{3}}{6} \right) = \frac{3}{4} (4\pi - 3\sqrt{3}) \text{ units}^2$ | 2                                  |
| (d)i)      |  <p> <math>PQ^2 = 360^2 + 480^2 - 2 \times 360 \times 480 \cos 110^\circ</math><br/> <math>PQ^2 = 478202</math><br/> <math>PQ = 692 \text{ km (nearest km)}</math> </p>                                                    | 2                                  |

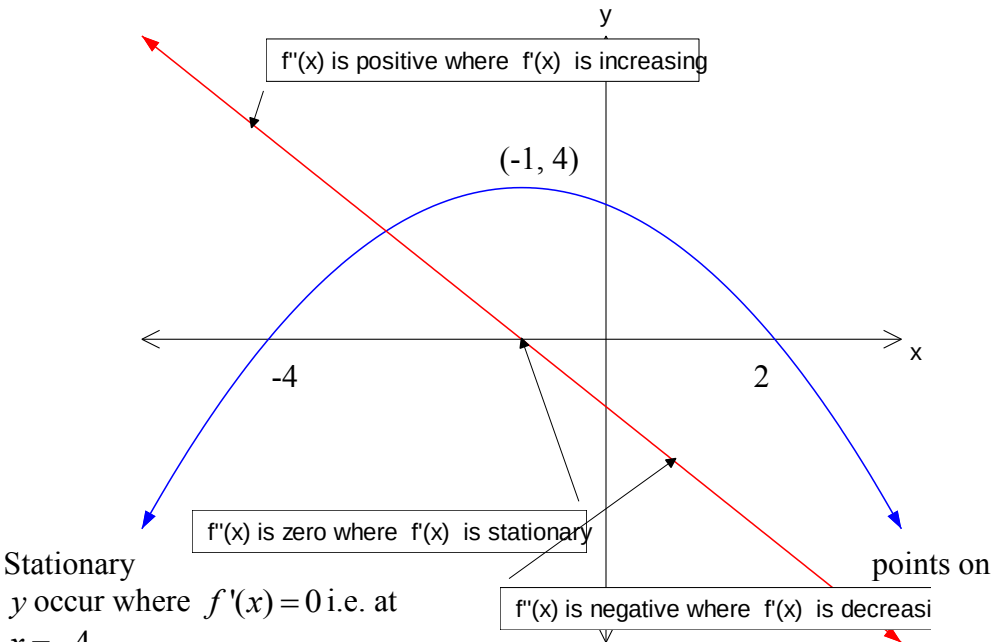
| Question 4 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Trial HSC Examination- Mathematics |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks                              |
| (d)ii      | <p>First find <math>\angle QPR</math></p> $\frac{\sin \angle QPR}{480} = \frac{\sin 110^\circ}{692}$ $\sin \angle QPR = \frac{480 \times \sin 110^\circ}{692}$ $\sin \angle QPR = 0.652$ $\angle QPR = 41^\circ$ $\angle NPR = 150^\circ$ $\text{Bearing}(\angle NPQ) = 150^\circ - 41^\circ$ $= 109^\circ$ <div style="border: 1px solid black; padding: 5px; margin-top: 10px; width: fit-content;"> <p>Can also be found using cos rule using the 3 sides..</p> </div> | 2                                  |

| Question 5 |                                                                                                                                                                                                                                                      | Trial HSC Examination- Mathematics |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                             | Marks                              |
| (a)i)      | $\angle EAD = 60^\circ$ (equilateral $\triangle EAD$ )<br>$\angle DAC = 45^\circ$ (isosceles right-angled $\triangle DAC$ )<br>$\therefore \angle EAB = \angle EAD + \angle DAC + \angle CAB$<br>$= 60^\circ + 45^\circ + 51^\circ$<br>$= 156^\circ$ | 3                                  |
| ii)        | $\angle ABC = 180^\circ - 156^\circ$ (interior $\angle$ on $\parallel$ lines, AE and BC)<br>$= 24^\circ$                                                                                                                                             | 1                                  |
| (b)i)      | $x = \frac{4t^2 + t + 8}{4t + 1}$ $x = \frac{4(0)^2 + (0) + 8}{4(0) + 1} \text{ when } t = 0$ $= 8$ $\therefore \text{initial displacement is 8 m}$                                                                                                  | 2                                  |
| (b)ii)     | $x = \frac{4t^2 + t + 8}{4t + 1}$ $\therefore \frac{(4t + 1)(8t + 1) - (4t^2 + t + 8)(4)}{(4t + 1)^2}$ $= \frac{32t^2 + 12t + 1 - 16t^2 - 4t - 32}{(4t + 1)^2}$ $= \frac{16t^2 + 8t - 31}{(4t + 1)^2}$                                               | 1                                  |

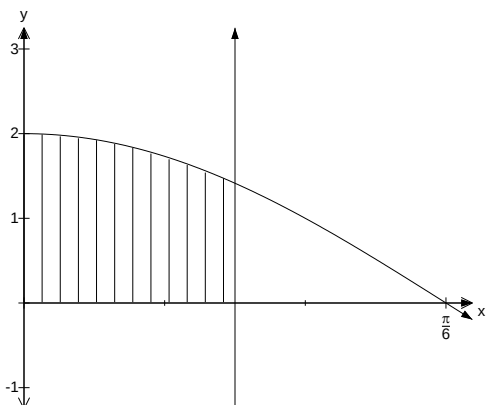
| Question 5 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Trial HSC Examination- Mathematics |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks                              |
| (b)iii)    | $\frac{16t^2 + 8t - 31}{(4t+1)^2} = 0$ $16t^2 + 8t - 31 = 0$ $t = \frac{-8 \pm \sqrt{8^2 - 4(16)(-31)}}{2(16)}$ $= \frac{-8 \pm \sqrt{2048}}{32}$ $= \frac{-8 \pm 32\sqrt{2}}{32}$ $= \frac{-1 \pm 4\sqrt{2}}{4}$ <p>So it is stationary when <math>t = \frac{-1 \pm 4\sqrt{2}}{4}</math></p> <p>Only use positive value so <math>t = \frac{-1 + 4\sqrt{2}}{4} \approx 1.2 \text{ s (to one dec pl)}</math></p>                                                                                                                                                                                                                                    | 2                                  |
| (b)iv)     | $t = \frac{-1 + 4\sqrt{2}}{4} \approx 1.2 \text{ s } v = 0, x \approx 2.58 \text{ (to 2 dec pl)}$ <p>When <math>t = 0</math> <math>x = 8</math><br/> When <math>t = 2</math> <math>x = 2.89</math> (to 2 dec pl)<br/> Particle starts 8 units to the right of the origin, moving toward the origin, it stops after 1.2 sec at 2.58 m to right of origin, then begins to move away from the origin, being 2.89 units to the right of the origin after 2 sec.<br/> <math>\therefore</math> total distance travelled is <math>(8 - 2.58) + (2.89 - 2.58)</math><br/> <math>= 5.73 \text{ m}</math> or <math>5.7 \text{ m}</math> (to 1 dec place)</p> | 2                                  |
| (c)        | $3x - y = 10 \quad (1)$ $x = y + 2 \quad (2)$ $3(y + 2) - y = 10 \quad (3) \text{ sub (2) in (1)}$ $3y + 6 - y = 10$ $2y = 4$ $y = 2$ $x = (2) + 2 \quad \text{sub } y \text{ in (2)}$ $x = 4$ <p>solution (4, 2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                              | 2                                  |

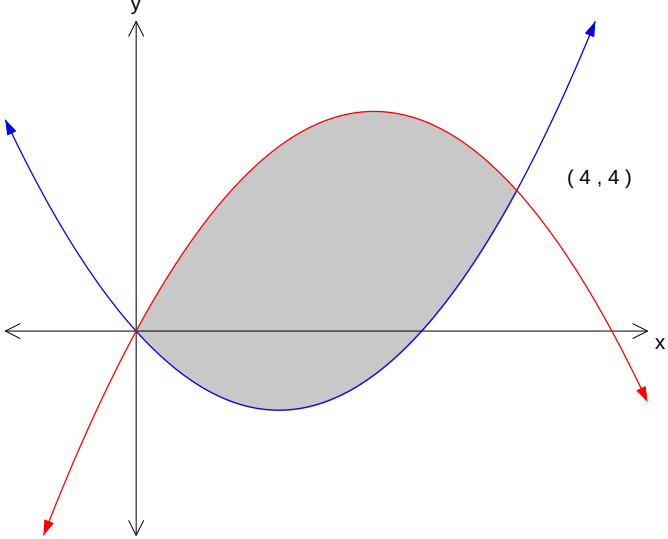
| Question 6 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Trial HSC Examination- Mathematics |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks                              |
| (a)(i)     | $y = x^6 - 6x^4$<br>Crosses axis where<br>$x^6 - 6x^4 = 0$<br>$x^4(x^2 - 6) = 0$<br>$x^4(x - \sqrt{6})(x + \sqrt{6}) = 0$<br>Crosses axis where $x = 0$ and $x = \pm\sqrt{6}$                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2                                  |
| (a)(ii)    | $y = x^6 - 6x^4$<br>$y' = 6x^5 - 24x^3$<br>$= 6x^3(x^2 - 4)$<br>$= 6x^3(x - 2)(x + 2)$<br>$y'' = 30x^4 - 72x^2$<br>Stationary points where<br>$x = 0, y = 0, y'' = 0$<br>$x = 2, y = -32, y'' = 192$<br>$x = -2, y = -32, y'' = 192$<br><br>Stationary points $(-2, -32), (0, 0), (2, -32)$<br>$y'' = 30x^4 - 72x^2$<br>At $x = 0$ $y'' = 0$ so test either side<br>At $x = 1$ $y'' = -42 \therefore$ concave down.<br>At $x = -1$ $y'' = -42 \therefore$ concave down<br>$\therefore$ maximum at $(0, 0)$<br>At $x = 2$ $y'' = 192 \therefore$ minimum at $(2, -32)$ .<br>At $x = -2$ $y'' = 192 \therefore$ minimum at $(-2, -32)$ . | 4                                  |

| Question 6 | Trial HSC Examination- Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
| (a)(iii)   | $y'' = 30x^4 - 72x^2$ $= 6x^2(5x^2 - 12)$ $= 6x^2(\sqrt{5}x - 2\sqrt{3})(\sqrt{5}x + 2\sqrt{3})$ $x = 0 \quad y = 0$ $x = \frac{2\sqrt{3}}{\sqrt{5}} = \frac{2\sqrt{15}}{5} \quad y = -20.736$ $x = -\frac{2\sqrt{3}}{\sqrt{5}} = -\frac{2\sqrt{15}}{5} \quad y = -20.736$ <p>Check for changes of concavity</p> <p>From above, no change at (0, 0) but</p> <p>there is a change at <math>\left(\pm \frac{2\sqrt{15}}{5}, -20.736\right)</math></p> <p>Inflexions at <math>\left(\pm \frac{2\sqrt{15}}{5}, -20.736\right)</math></p> | 2     |
|            | <p>Graph of the function <math>y = 30x^4 - 72x^2</math>. The graph shows a local maximum at <math>(0, 0)</math> and two local minima at <math>(-2, -32)</math> and <math>(2, -32)</math>. The inflection points are marked at <math>\left(-\frac{2\sqrt{15}}{5}, -20.7\right)</math> and <math>\left(\frac{2\sqrt{15}}{5}, -20.7\right)</math>.</p>                                                                                                                                                                                  | 2     |

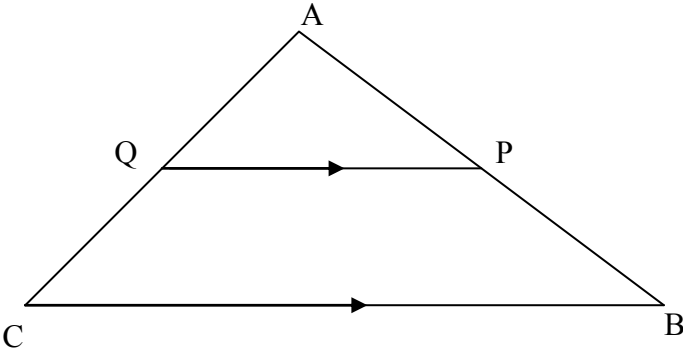
| Question 6 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Trial HSC Examination- Mathematics |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks                              |
| (b)        |  <p>Stationary<br/>y occur where <math>f'(x) = 0</math> i.e. at<br/><math>x = -4</math><br/>here <math>f''(x)</math> is positive <math>\therefore</math> min turning point at <math>x = -4</math><br/>and at <math>x = 2</math><br/>here <math>f''(x)</math> is negative <math>\therefore</math> max turning point at <math>x = 2</math><br/>or using first derivative test (explanation or table)</p> | 2                                  |

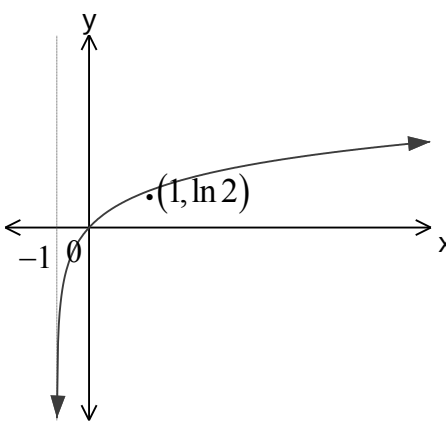
| Question 7 |                                                                         | Trial HSC Examination- Mathematics |
|------------|-------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                | Marks                              |
| (a)i)      | $\log_a 0.4 = \log_a \frac{2}{5}$ $= \log_a 2 - \log_a 5$ $= x - y$     | 1                                  |
| ii)        | $\log_a 20 = \log_a (2^2 \times 5)$ $= 2\log_a 2 + \log_a 5$ $= 2x + y$ | 2                                  |

| Question 7 |                                                                                                                                                                                                                                                                                                                                                                                                                         | Trial HSC Examination- Mathematics |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                                                                                                                | Marks                              |
| b)         | <p><math>y = 2 \cos 3x</math></p>  $  \begin{aligned}  \text{Area} &= \int_0^{\frac{\pi}{12}} 2 \cos 3x \, dx \\  &= \left[ \frac{2 \sin 3x}{3} \right]_0^{\frac{\pi}{12}} \\  &= \frac{2 \sin \frac{\pi}{4}}{3} - \frac{2 \sin 0}{3} \\  &= \frac{2}{3\sqrt{2}} - 0 \\  &= \frac{\sqrt{2}}{3} \text{ square units.}  \end{aligned}  $ | 3                                  |
| (c)i)      | <p>Substitute <math>y = x^2 - 3x</math> into <math>y = 5x - x^2</math></p> $5x - x^2 = x^2 - 3x$ $2x^2 - 8x = 0$ $2x(x - 4) = 0$ $x = 0, \quad y = 0$ $x = 4, \quad y = 4$ <p>Intersect at (0, 0) and (4, 4).</p>                                                                                                                                                                                                       | 2                                  |

|     |                                                                                                                                                                                                                                                                                                                                                  |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| ii) |  $  \begin{aligned}  \text{Area} &= \int_0^4 5x - x^2 dx - \int_0^4 x^2 - 3x dx \\  &= \int_0^4 8x - 2x^2 dx \\  &= \left[ 4x^2 - \frac{2x^3}{3} \right]_0^4 \\  &= \left( 64 - \frac{128}{3} \right) - 0 \\  &= \frac{64}{3} = 21\frac{1}{3}  \end{aligned}  $ | 3 |
| d)  | $  \begin{aligned}  &\frac{d}{dx} \log_e (\cos x) \\  &= \frac{-\sin x}{\cos x} \\  &= -\tan x  \end{aligned}  $                                                                                                                                                                                                                                 |   |

| Question 8 |                                                                                                                                                                                                                                                                                                                                                    | Trial HSC Examination- Mathematics |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                                           | Marks                              |
| (a)i)      | $P = Ae^{kt}$ $\frac{dP}{dt} = Ae^{kt} \cdot k$ $= kAe^{kt}$ $= kP$                                                                                                                                                                                                                                                                                | 1                                  |
| ii)        | $t = 1 \text{ was } 147\,200$ $P = Ae^{kt}$ $147200 = Ae^k \quad \text{(i)}$ $t = 2 \text{ was } 154\,800$ $154800 = Ae^{2k} \quad \text{(ii)}$ $\frac{154800}{147200} = \frac{Ae^{2k}}{Ae^k} \quad \text{(ii) } \div \text{(i)}$ $1.0516 = e^k$ $k = \ln(1.0516)$ $k \approx 0.05$ $147200 = Ae^{0.05(1)}$ $A = \frac{147200}{e^{0.05}} = 139973$ | 2                                  |
| iii)       | <p>When <math>t = 4</math></p> $P = Ae^{kt}$ $P = 139973e^{0.05(4)}$ $= 171197$                                                                                                                                                                                                                                                                    | 1                                  |
| iv)        | $P = Ae^{kt}$ $200000 = 139973e^{0.05t}$ $\frac{200000}{139973} = e^{0.05t}$ $1.429 = e^{0.05t}$ $\ln(1.429) = \ln(e^{0.05t})$ $0.05t = \ln(1.429)$ $t = \frac{\ln(1.429)}{0.05}$ $= 7.1$ <p><math>t = 7</math> is start of 2012<br/>Population will reach 200 000 in 2012</p>                                                                     | 1                                  |

| Question 8 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Trial HSC Examination- Mathematics |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks                              |
| (b)i)      |  <p>In <math>\triangle APQ</math> and <math>\triangle ABC</math><br/> <math>\angle A</math> is common<br/> <math>\angle AQP = \angle ACB</math> (Corresponding <math>\angle</math>s on <math>\parallel</math> lines)<br/> <math>\angle APQ = \angle ABC</math> (Corresponding <math>\angle</math> on <math>\parallel</math> lines)<br/> <math>\therefore \triangle APQ \parallel \triangle ABC</math> (Corresponding angles equal)</p> | 2                                  |
| ii)        | $\frac{AP}{AB} = \frac{1}{2} \quad (\text{P is midpoint of AB})$ $\frac{AP}{AB} = \frac{AQ}{AC} \quad (\text{sides of similar triangle in same ratio})$ $\frac{AQ}{AC} = \frac{1}{2} \quad (\text{from above})$ $\therefore Q \text{ is midpoint of AC.}$                                                                                                                                                                                                                                                                | 2                                  |
| (c)        | $\int_1^3 g(x) dx \approx \frac{1}{6} \{12 + 4(8) + 2(0) + 4(3) + 5\}$ $\approx \frac{61}{6}$ $\approx 10\frac{1}{6}$                                                                                                                                                                                                                                                                                                                                                                                                    | 2                                  |
| (d)        | $\sum_{n=2}^5 (n^2 - 1) = (2^2 - 1) + (3^2 - 1) + (4^2 - 1) + (5^2 - 1)$ $= 3 + 8 + 15 + 24$ $= 50$                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                  |

| Question 9 |                                                                                                                                                                                                            | Trial HSC Examination- Mathematics |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                   | Marks                              |
| (a)        | $y = 2x^2 - 2$ $V = \pi \int_0^6 x^2 dy$ $= \pi \int_0^6 \frac{y+2}{2} dy$ $= \pi \left[ \frac{y^2}{4} + y \right]_0^6$ $= \pi \left[ \left( \frac{36}{4} + 6 \right) - (0) \right]$ $= 15\pi \text{ u}^3$ | 3                                  |
| (b)        |                                                                                                                          | 2                                  |
| (c)        | $\frac{2}{5} \int 5x^4 e^{x^5} dx = \frac{2}{5} e^{x^5} + c$                                                                                                                                               | 1                                  |
| (c)i)      | $A_6 = 20000 - 6M$                                                                                                                                                                                         | 1                                  |
| ii)        | $A_7 = (20000 - 6M)1.01 - M$ $A_8 = [(20000 - 6M)1.01 - M]1.01 - M$ $= (20000 - 6M)1.01^2 - 1.01M - M$ $= (20000 - 6M)1.01^2 - M(1 + 1.01)$                                                                | 1                                  |
| iii)       | $A_9 = (20000 - 6M)1.01^3 - M(1 + 1.01 + 1.01^2)$ $A_n = (20000 - 6M)1.01^{n-6} - M(1 + 1.01 + 1.01^2 + \dots + 1.01^{n-7})$ $A_{36} = (20000 - 6M)1.01^{30} - M(1 + 1.01 + 1.01^2 + \dots + 1.01^{29})$   | 2                                  |

| Question 9 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Trial HSC Examination- Mathematics |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part       | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks                              |
| iv)        | <p>Since repaid after 36 months <math>A_{36} = 0</math></p> $(20000 - 6M)1.01^{30} - M(1 + 1.01 + 1.01^2 + \dots + 1.01^{29}) = 0$ $M(1 + 1.01 + 1.01^2 + \dots + 1.01^{29}) = (20000 - 6M)1.01^{30}$ <p>Need to evaluate <math>1 + 1.01 + 1.01^2 + \dots + 1.01^{29}</math></p> <p>Geometric series with <math>a = 1</math>, <math>r = 1.01</math>, <math>n = 30</math></p> $S_{30} = \frac{a(r^n - 1)}{r - 1}$ $= \frac{1(1.01^{30} - 1)}{1.01 - 1}$ $= 34.785$ $34.785M = (20000 - 6M)1.01^{30}$ $\frac{34.785M}{1.01^{30}} = 20000 - 6M$ $6M + \frac{34.785M}{1.01^{30}} = 20000$ $M\left(6 + \frac{34.785}{1.01^{30}}\right) = 20000$ $31.8M = 20000$ $M = \frac{20000}{31.8}$ $= \$629 \quad (\text{nearest dollar})$ | 2                                  |

| Question 10 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Trial HSC Examination- Mathematics |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Part        | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks                              |
| a) i)       | $V = 2 - \sqrt{3} \cos t - \sin t$ $\frac{dV}{dt} = \sqrt{3} \sin t - \cos t$                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                                  |
| ii)         | <p>When <math>t = 0</math></p> $\frac{dV}{dt} = \sqrt{3} \sin 0 - \cos 0$ $= -1$ <p><math>\therefore</math> the tank is emptying at this time.</p>                                                                                                                                                                                                                                                                                                                                                                               | 1                                  |
| iii)        | <p>Full (or empty) when <math>\frac{dV}{dt} = 0</math></p> $\frac{dV}{dt} = 0$ $\sqrt{3} \sin t - \cos t = 0$ $\sqrt{3} \sin t = \cos t$ $\frac{\sin t}{\cos t} = \frac{1}{\sqrt{3}}$ $\tan t = \frac{1}{\sqrt{3}}$ $= \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6} \dots$ <p>As tank is initially emptying,<br/>second value corresponds to when full.</p> <p>Tank is first full when <math>t = \frac{7\pi}{6}</math></p> $V = 2 - \sqrt{3} \cos t - \sin t$ $= 2 - \sqrt{3} \cos \frac{7\pi}{6} - \sin \frac{7\pi}{6}$ $= 4$ | 3                                  |

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
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| (b)i)  | $\angle D$ is common<br>$\angle C = \angle B = 90^\circ$<br>$\angle E = \angle A$ (corresponding angles on    lines)<br>$\triangle ABD \parallel \triangle ECD$ (equiangular)<br>$\frac{x+h}{x} = \frac{2t}{2} = t$<br>$x+h = tx$<br>$tx - x = h$<br>$x(t-1) = h$<br>$x = \frac{h}{t-1}$                                                                                                                                                                                                           | 2 |
| (b)ii) | $V = \frac{1}{3}\pi(2t)^2 \cdot (h+x) - \frac{1}{3}\pi(2)^2 \cdot x$ $= \frac{1}{3}\pi(2t)^2 \cdot \left(h + \frac{h}{t-1}\right) - \frac{1}{3}\pi(2)^2 \cdot \left(\frac{h}{t-1}\right)$ $= \frac{1}{3}\pi(2t)^2 \cdot \left(\frac{ht}{t-1}\right) - \frac{1}{3}\pi(2)^2 \cdot \left(\frac{h}{t-1}\right)$ $= \frac{1}{3}\pi(2)^2 \cdot \left(\frac{h}{t-1}\right)(t^3 - 1)$ $= \frac{4}{3}\pi \cdot \left(\frac{h}{t-1}\right)(t-1)(t^2 + t + 1)$ $= \left(\frac{4\pi h}{3}\right)(t^2 + t + 1)$ | 2 |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| iii) | <p>Sum of radii and height = 12<br/> <math>2 + h + 2t = 12</math><br/> <math>h = 10 - 2t</math><br/> <math>V = \left(\frac{4\pi h}{3}\right)(t^2 + t + 1)</math><br/> <math>= \left(\frac{4\pi}{3}\right)(10 - 2t)(t^2 + t + 1)</math><br/> <math>= \left(\frac{4\pi}{3}\right)(10t^2 + 10t + 10 - 2t^3 - 2t^2 - 2t)</math><br/> <math>= \left(\frac{4\pi}{3}\right)(8t^2 + 8t - 2t^3 + 10)</math><br/> <math>V = \left(\frac{4\pi}{3}\right)(8t^2 + 8t - 2t^3 + 10)</math><br/> <math>\frac{dV}{dt} = \left(\frac{4\pi}{3}\right)(16t + 8 - 6t^2)</math><br/> <math>\frac{dV}{dt} = 0</math><br/> <math>(16t + 8 - 6t^2) = 0</math><br/> <math>6t^2 - 16t - 8 = 0</math><br/> <math>3t^2 - 8t - 4 = 0</math><br/> <math>t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}</math><br/> <math>= \frac{8 \pm \sqrt{(-8)^2 - 4(3)(-4)}}{2(3)}</math><br/> <math>= \frac{8 \pm \sqrt{112}}{6}</math><br/> <math>= -0.43 \text{ or } 3.10 \text{ (to 2 dec pl)}</math><br/> <math>t &gt; 0</math> so <math>t = 3.10 \text{ m}</math><br/> <math>\frac{d^2V}{dt^2} = \left(\frac{4\pi}{3}\right)(16 - 12t)</math><br/> <math>\frac{d^2V}{dt^2} = \left(\frac{4\pi}{3}\right)(16 - 12(3.10)) = \left(\frac{4\pi}{3}\right)(-21.2)</math><br/> <math>= -88.7</math><br/> When <math>t = 3.10</math> <math>\therefore \frac{d^2V}{dt^2} &lt; 0</math><br/> <math>\therefore V</math> is a maximum<br/> <math>V = \left(\frac{4\pi}{3}\right)(8(3.10)^2 + 8(3.10) - 2(3.10)^3 + 10)</math><br/> <math>= 218.2 \text{ m}^3 \text{ (to 1 dec pl)}</math></p> | 3 |
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